

Quantum Fault Tolerance Meets Toom's Contours

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1 Computation Model

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Separate between an element gate $g \in \mathcal{G}$ and a gate, which also has a space-time location.

Definition 1.1 (Quantum Circuit). A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g: L(\mathcal{H}_2^{\otimes \Delta}) \rightarrow L(\mathcal{H}_2^{\otimes \Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C , with depth d and width w , is a collection of tuples $\{u_i = (g_i, l_i)\}_{i=1}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^\Delta$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t -cut of C to be $C|_t := \{u_i = (g_i, l_i): u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the directed acyclic graph associated with C .

Definition 1.2 (Associated Computation DAG). Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w . We define the associated computation DAG of C , denoted by π_C , to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C , namely $\mathcal{U} = \{u_i\}_{i=1}$.
2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:

$$j \in \arg \min_k \{l_{k,1} : (g_k, l_k) \in \mathcal{U}, l_{k,1} > l_{i,1} \text{ and } l_{k,2} \cap l_{i,2} \neq \emptyset\}$$

Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C . Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}_{i=1}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$
2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$. We name $C_{J,i}$ the partial computation circuit until step i .

Definition 1.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w . For any density matrix σ , quantum gate $g \in \mathcal{G}$, and location l , the application of the gate-location tuple (g, l) on σ , denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ , denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ , according to a topological order induced by π_C .

We say that $\rho_1, \rho_2, \dots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u'_i\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \dots, \rho_k$ a running.

Definition 1.4 (C Subjected to $\mathcal{E}$ at frequency r). Let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w . For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, and a number $r \in (0, 1)$ satisfies $r^{-1} \in \mathbb{N}$, we define the quantum circuit C subjected to $\mathcal{E}$ at frequency r , denoted by $C[\mathcal{E}]_r = \{u'_i\}_{i=1}$, to be the quantum circuit that is the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) = \begin{cases} (g_i, (l_{i,1} + \lfloor r l_{i,1} \rfloor + 1, l_{i,2})) & u_i \in C \\ (g_i, (r^{-1} l_{i,1}, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

Where r is not mentioned, we mean $r = \frac{1}{2}$, and the result of the subsection is the circuit whose gates on the odd time coordinates are the gates of C , and whose gates on the even time coordinates are the gates of $\mathcal{E}$. We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 1.5 (Faults Propagation to a Step τ). For a quantum circuit $C = \{u_i\}_{i=1}$, with depth d , an order $J = \{j_1, j_2, \dots, j_k\} \subset [|C|]$ that agrees with π_C , and gate u_i in C , a *propagation of a gate* $u_i \in C$ to step τ is the quantum operator $X: L(\mathcal{H}_2^w) \rightarrow L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J . If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \leq \tau - l$, look for an operator $X_k: L(\mathcal{H}_2^w) \rightarrow L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ .

We define the *propagated computation graph* $(\mathcal{U}', E')$, resulted by the propagation of u_i to step τ , as follow: Let $u_i = u_{j_1}, \dots, u_{j_l}$, be the gates at steps between i to τ , and let X be the propagation of u_i to step τ . Consider the tree rooted at u_i restricted to $u_{j_1}, \dots, u_{j_l}$, denote it by T and it leaves by L_T . Set $\mathcal{U}' = \mathcal{U}/u_i \cup X$ and:

$$\begin{aligned} E' = & E \\ & / \quad \{ \{u, u_i\} : \{u, u_i\} \in E \} \\ & / \quad \{ \{u, v\} : u \in L_T, \{u, v\} \in E \} \\ & \cup \quad \{ \{X, v\} : u \in L_T, \{u, v\} \in E \} \\ & \cup \quad \{ \{u, X\} : u \in L_T \} \\ & \cup \quad \{ \{u, v\} : \{u, u_i\}, \{u_i, v\} \in E \} \end{aligned}$$

For a quantum circuit C , a set of faults $\mathcal{E} = \{e_i\}_{i=1}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X_{e_i}^\tau$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ . Let $j'_1, j'_2, \dots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ , ordered according to the reverse topological order induced by $\pi_{\mathcal{E}}$. The propagated error to step τ is defined by

$$X^\tau := \prod_i X_{e_{j'_i}}^\tau$$

Definition 1.6 (Deviation). For a quantum circuit C , a set of faults $\mathcal{E}$, and step τ , let X^τ be the propagated error at step τ . We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 1.7. Using the same notations as in Definition 1.6, for a set of unitary operators $\mathcal{S} = \{s_i : \mathcal{H}_2^w \rightarrow \mathcal{H}_2^w\}_{i=1}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}} \{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

The following, extends the notion of Tanner graphs to the context of quantum circuits.

Definition 1.8 (Tanner Graph of a Register). For a quantum circuit C , a register R is a subset of qubits in C . The *Tanner graph* at step τ of register R , denoted by $\mathcal{T}_{R,\tau}$, is the graph defined as follows:

1. If there is a code $\mathcal{C}$, such that at turn τ , the qubits in C are an encoding of $\mathcal{C}$, then $\mathcal{T}_{R,\tau}$ is the Tanner graph of $\mathcal{C}$.
2. Otherwise, $\mathcal{T}_{R,\tau}$ is the empty graph.

Where we consider a range of turns in which R is encoded in the same code, and the turn is clear from the context, we will be brief by omitting the turn subscript.

Definition 1.9 (Generalized Tanner Graph of a Circuit). For a quantum circuit C , and registers a and b in C , the *generalized Tanner graph* of the register obtained by taking their union $a \cup b$ is $\mathcal{T}_{a \cup b} := \mathcal{T}_a \cup \mathcal{T}_b$. For a partition of the qubits into registers $\{R_i\}_{i=1}$, the *generalized Tanner graph* of a quantum circuit C , at step τ , according to the partition $\{R_i\}_{i=1}$, is the graph obtained by:

$$\mathcal{T}_{C, \{R_i\}_{i=1}, \tau} := \bigcup_{i=1} \mathcal{T}_{R_i, \tau}$$

When the partition is clear, we will brief and use $\mathcal{T}_C$ to define the generalized Tanner graph of C .

Definition 1.10 (Clustered Deviation). For a quantum circuit C , a register R of C , and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R , and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ . Denote by $\mathcal{T}_R|_{\mathcal{D}_\tau}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_\tau$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_\tau}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 1.11. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2 . Fix a set of faults $\mathcal{E}$, and an order J that agrees with $\pi_{C[\mathcal{E}]}$. For any τ , denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \dots, Y_l^{(\tau)}$ the deviation clusters at step $\tau \in J$. Assume that at step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}|$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z , such $Z \cap Y_i^{(\tau_1)} \neq \emptyset$ and Z is of size at least:

$$|Z| \geq \frac{1}{\Delta \Delta_1 \Delta_2} |Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z' , such $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$, and Z' is of size at most:

$$|Z'| \leq \Delta \Delta_1 \Delta_2 |Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time t can be decomposed into a product $\Rightarrow$ at each step, a fault is supported on at most one (q)bit.

Proof. Denote by $u = (g, l)$ the gate at step τ . If u acts trivially on the (q)bits in $Y_i^{(\tau_1)}$ then they also belong to deviation on register R at step $\tau_1 - 1$, Since they are connected in $\mathcal{T}$ they form a deviation cluster on register R at step $\tau_1 - 1$ and we got the desired. Thus it is left to handle the case in which u acts non-trivially on (q)bits in $Y_i^{(\tau_1)}$. Since u acts on at most Δ qubits, and each qubit has at most $\Delta_1(\Delta_2 - 1)$ neighbors at $\mathcal{T}$ it follows that u acts non trivially on at most

$$\Delta + \Delta \Delta_1 (\Delta_2 - 1) \leq \Delta \Delta_1 \Delta_2$$

deviation clusters at step $\tau_1 - 1$. Let l be integer no greater than $\Delta\Delta_1\Delta_2$ and let $Y_1, Y_2, \dots, Y_l$ be the deviation cluster, on register R , at step $\tau_1 - 1$, on which u acts non trivially, since $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$. We have that:

$$\begin{aligned} |Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2} |Y_i^{(\tau_1)}| \end{aligned} \tag{1}$$

Complete the second inequality. Should consider to erase what have already written about it. To get the second inequality, we use the fact that u is a unitary gate, in particular, u is reversible. Let $C_{J,\tau_1} = u_1, \dots, u_{\tau_1}$ be the partial computation circuit until step τ_1 . Observe that C_{J,τ_1-1} is obtained by stacking $u_{\tau_1}^\dagger$ as a final stage to C_{J,τ_1} , denoted by $u^\dagger \circ C_{J,\tau_1}$. Thus, the deviation from C_{J,τ_1} at step $\tau_1 - 1$ equals the deviation from $u^\dagger \circ C_{J,\tau_1}$ at step $\tau_1 + 1$. By eq. (1), $\square$

Corollary 1.12 (Intermediate Value Theorem). Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$, and let x be an integer. Assume that at the initial step τ_o , there is no deviation cluster of size more than αx , and that at a step $\tau_1 \in J$, such that $\tau_1 > \tau_o$, there is a deviation cluster $Y^{(\tau_1)}$ of size $|Y^{(\tau_1)}| > x$. Then there is a step $\tau_\star \in J$ such that $\tau_o < \tau_\star < \tau_1$ for which there is a deviation cluster Z at step $\tau_\star$, satisfying that the size of Z is in the range $(\alpha x, x)$.

Proof. Assume through contradiction that for any turn $\tau \in (\tau_o, \tau_1)$, all the deviation clusters are of size smaller than αx . Particularly, at turn $\tau_1 - 1$, all the deviation clusters are of size smaller than αx . Yet, by Claim 1.11, the existence of the deviation cluster $Y^{(\tau_1)}$, at size d , implies the existence of a deviation cluster at turn $\tau_1 - 1$, of size at least αx , which leads to a contradiction. $\square$

Claim 1.13. Let C be a quantum circuit with order J and depth d , and let $\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}$, so the circuit $C[\mathcal{E}]$ is a random variable, and it is fan-bounded by Δ . For a time $t \leq 2d$ and an integer $x \in \mathbb{N}$, let A be the event in which, until time t , there is a time $t' \leq t$ such that there is a deviation cluster of size greater than x . For any step $\tau \in J$, let B_τ be the event that step τ is the first step to have a deviation cluster of size in the range $(\alpha x, x)$. Then:

$$\Pr[A] \leq 2wd \max_{\tau \in [2wd]} \Pr[B_\tau]$$

Proof. By Corollary 1.12, we have that the occurrence of a deviation cluster of size greater than x before time t implies that there is a step $\tau \in J$, such that $\tau \leq 2wt$, in which there is a deviation cluster of size greater than αx . Thus,

$$A \subset \bigcup_{\tau \in [2wt]} B_\tau \subset \bigcup_{\tau \in [2wd]} B_\tau$$

So by using the union bound we get:

$$\begin{aligned} \Pr[A] &\leq \Pr \left[\bigcup_{\tau \in [2wd]} B_\tau \right] \leq \sum_{\tau \in [2wd]} \Pr[B_\tau] \\ &\leq 2wd \max_{\tau \in [2wd]} \Pr[B_\tau] \end{aligned}$$

□

[COMMENT] Separate the above from the beneath. Explain that we introduce two notions of propagation on purpose.

Definition 1.14 (Layer Propagation.). Let $C = \{u_i\}_{i=1}$ be a quantum circuit, and let $U_1, U_2, \dots, U_d$ be a partition of the gates in C according to their time coordination. We name such a partition the *partition of C into layers*. The *forward layer propagation* of U_i is defined to be the action that results in the quantum circuit such that its partition into layers is $U_1, \dots, U_{i-1}, U_{i+1}, V, U_{i+2}, \dots, U_t$, where V is a collection of quantum gates, such that

$$\prod_{u \in U_i} u \prod_{u \in U_{i+1}} u = \prod_{u \in U_{i+1}} u \prod_{u \in V} u$$

For $i' > i$, the *forward layer propagation* of U_i to time i' is defined by propagating U_i forward for $i' - i$ times. In the resulting circuits of the layer propagation actions, we update the time coordinates to match their application turn.

Claim 1.15. The result of forward layer propagation has a bounded fan-in.

[COMMENT] Prove it.

Definition 1.16 ($C \sim_P C'$). We define the equivalence relation over quantum circuits $\sim_P$ to be satisfied if any of them can be obtained from the other by propagating layers. Namely, for quantum circuits C, C' , such that the partition of C into layers is $U_1, \dots, U_d$, we have $C \sim_P C'$ if there is a subset of layers $\{U_i\}_{i=1} \subset \{U_1, \dots, U_d\}$ and subsets of time points $\{t_i\}_{i=1}$ such that C' is the result of propagating each of the U_i to time t_i in C according to the order indexed by i .

Note that, by definition, we find that equivalence between quantum circuits by $\sim_P$ implies an equivalence of their action. More specifically, for a mixed state ρ and quantum circuits C and C' such that $C \sim_P C'$, we have:

$$C\rho C^\dagger = C'\rho C'^\dagger$$

Definition 1.17 (Noise Channel and The Local Stochastic Noise Property). A *noise channel* $\mathcal{N}_{\mathcal{E}}$ is a distribution over sets of faults, we will denote the outcome of $\mathcal{N}_{\mathcal{E}}$ by $\mathcal{E}$. For $p \in (0, 1)$ we say that a noise channel $\mathcal{N}_{\mathcal{E}}$ is a *local stochastic*, with noise rate p , if the following holds: For every subset of locations $S \subset \mathbb{N} \times \mathbb{N}$, whose locations have the same time coordinate, the probability that S is covered by $\mathcal{E}$ is less than:

$$\Pr_{\mathcal{E} \sim \mathcal{N}_{\mathcal{E}}} [S \subset \mathcal{E}] \leq p^{|S|}$$

Observes that a quantum circuit C and a noise channel $\mathcal{N}_{\mathcal{E}}$ induce a subsection $C[\mathcal{E}]$, which is a random variable.

Claim 1.18. Let $\mathcal{N}_{\mathcal{E}}$ be a local stochastic noise channel with noise rate p , let C be a quantum circuit with bounded fan-in Δ , and let c be an integer. We define q to be the number that depends on p , Δ , and c as: $q := (\Delta^c p)^{\frac{1}{\Delta^c}}$. There is a local stochastic noise channel $\mathcal{N}_{\mathcal{E}_q}$, with noise rate q that induces a subjection $C[\mathcal{E}_q]_{\frac{1}{c}}$ such that:

$$C[\mathcal{E}]_{\frac{1}{2}} \sim_P C[\mathcal{E}_q]_{\frac{1}{c}}$$

Proof. We need to show that any subjection outcome $C[\mathcal{E}]$ can be transformed by propagation to a circuit C' that is also a result of subjecting C to a set of faults $\mathcal{E}'$ at frequency $\frac{1}{c}$, and that $\mathcal{E}'$ satisfies the local noise property with a noise rate q . Assume that c is even, and consider the layer propagations:

$$\begin{aligned} U_0, U_1, U_2, \dots, U_{c-1}, U_c, U_{c+1}, \dots, U_c &\mapsto \\ U_0, U_1, U_3, U_5, \dots, U_{c-1}, V_2, V_4, \dots, V_{c-2}, U_c, U_{c+1}, U_{c+3}, \dots &\mapsto \\ V^{(0)}, U_1, U_3, U_5, \dots, U_{c-1}, V^{(1)}, U_{c+1}, U_{c+3}, \dots & \end{aligned}$$

Where we grouped V_2, V_4, V_6, V_{c-2} together into $V^{(i)}$. Now, $\mathcal{E}' = \{V^{(i)}\}_{i=1}$, and $\square$

For that we need the follow property, for every subset of qubits S , there is subset of qubits S' at size at least $(1 - \gamma)|S|$ such that for S to be deviated, all the qubits in S' have to absorb fault. Now suppose that the gate is at depth at most Δ , and the fanning is at most c_f , then we get:

$$\begin{aligned} \Pr[S \in \mathcal{D}] &\leq (c_f^\Delta p)^{|S'|} \leq (c_f^\Delta p)^{(1-\gamma)|S|} \\ \Rightarrow p &\rightarrow (c_f^\Delta p)^{1-\gamma} \end{aligned}$$

A trivial lower bound for $1 - \gamma$ is .

$$\binom{\Lambda(S)}{|S'|} p^{|S'|} \sim 2^{|S|h(\alpha)} p^{|S'|}$$

[COMMENT] Denote by $\Lambda(S)$ the light cone of S . We need to iterate over all the minimal subsets of qubits that have to be flipped to make all the qubits in S flipped. We bound this number from above by the binomial coefficient, and then by the entropy function. We have to add a definition of minimal subset. In addition, bring all the definitions about layer propagations here, and add a paragraph that explains that we separate the two notations on purpose, as each one of them is defined to prove a different theorem.